

The Impact of the Brexit Depreciation Shock on the UK Labor Market: A Structural VAR Analysis

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Abstract

This paper investigates the dynamic macroeconomic effects of the Sterling depreciation following the 2016 Brexit referendum, with a specific focus on the labor market. Using a Structural Vector Autoregression (SVAR) model identified via *Mixed Sign Restrictions* on monthly data from 2000 to 2024, I isolate the specific exchange rate shock. My findings indicate that the "Uncertainty Channel" (Demand Shock) dominated the "Labor Supply Channel" (Supply Shock). Specifically, I observe a statistically significant and persistent increase in unemployment, suggesting that the contractionary effects of uncertainty and imported inflation outweighed the theoretical tightness of the labor market due to reduced migration. These results hold despite the immediate accommodative monetary policy response by the Bank of England.

1 Introduction

The outcome of the European Union referendum on June 23, 2016 ("Brexit"), represents one of the most significant structural breaks in the recent economic history of the United Kingdom. The immediate aftermath of the vote was characterized by a sharp and persistent depreciation of the British Pound (Sterling), alongside a surge in economic and political uncertainty.

While the financial reaction was instantaneous, the transmission of this shock to the real economy—and particularly to the labor market—remains a subject of debate. This project aims to quantify the impact of the Brexit-induced depreciation shock using a structural econometric approach, addressing a fundamental dichotomy regarding the nature of the crisis:

- **The Labor Supply Hypothesis:** The anticipation of stricter immigration controls and the exit from the Single Market could be interpreted as a negative labor supply shock. Theoretically, a reduction in the supply of EU workers should tighten the labor market, potentially lowering unemployment (or keeping it low) and putting upward pressure on wages, *ceteris paribus*.

- **The Uncertainty (Demand) Hypothesis:** Conversely, the uncertainty regarding future trade arrangements and the sudden rise in import costs (imported inflation) may have acted as a negative demand shock. In this scenario, firms would freeze investment and hiring plans due to a "wait-and-see" approach, leading to an increase in unemployment.

By employing a **Structural Vector Autoregression (SVAR)** model, I aim to disentangle these effects. Standard recursive identification methods (i.e., Cholesky decomposition) often fail to capture the simultaneous reaction of financial variables and monetary policy during crisis events. Therefore, I adopt an identification strategy based on **Mixed Sign Restrictions** (combining zero and sign constraints). This allows me to model the simultaneous fall in the exchange rate and the policy rate cut by the Bank of England, while assuming a sluggish response of real variables.

My empirical results provide evidence in support of the *Uncertainty Hypothesis*, showing that the depreciation shock generated a persistent rise in unemployment, overcoming the potential labor supply constraints.

2 Data and Identification Strategy

2.1 Dataset Description

I estimate the SVAR model using monthly data for the United Kingdom, covering the period from **January 2000 to March 2024** ($T \approx 290$ observations). This sample size captures the pre-crisis period, the Great Recession, and the entire post-referendum era, providing a comprehensive view of the UK economy under an inflation-targeting regime.

The vector of endogenous variables, y_t , includes six key macroeconomic indicators, ordered as follows:

1. **Brent Oil Price (Oil):** Included as a proxy for global supply shocks. Transformed into log-differences (growth rate).
2. **Industrial Production (IP):** A monthly measure of real economic activity. Transformed into log-differences.
3. **CPI Inflation (CPI):** The annual inflation rate calculated as the log-difference of the Consumer Price Index.
4. **Unemployment Rate (U):** The main variable of interest for testing the labor market response. Included in levels.
5. **Official Bank Rate (R):** The main monetary policy instrument of the Bank of England. Included in levels. Although the central bank implemented unconventional measures (QE) during parts of the sample, the Official Bank Rate remained the primary signaling tool for conventional policy adjustments and remained positive throughout the period (reaching a minimum of 0.10%).

6. **Real Effective Exchange Rate (REER):** The trade-weighted exchange rate of Sterling. An increase indicates appreciation, while a decrease indicates depreciation. Included in levels.

2.2 Lag Selection

The selection of the autoregressive lag order (p) is a critical step to ensure the model captures the dynamic interactions between variables without over-parametrization.

To determine the optimal lag length, I rely on the **Akaike Information Criterion (AIC)**, which minimizes at $p = 4$. This specification allows the model to capture the relevant short-term dynamics while maintaining a parsimonious structure, ensuring efficient estimation and preserving degrees of freedom.

3 Reduced-Form Estimation and Inference

3.1 Reduced-Form VAR

Before proceeding to the structural identification, I estimate the reduced-form Vector Autoregression (VAR) model. The dynamic system can be represented as:

$$y_t = c + \sum_{j=1}^p A_j y_{t-j} + u_t \quad (1)$$

where y_t is the $(n \times 1)$ vector of endogenous variables described in the previous section, c is a vector of constants, A_j are the $(n \times n)$ matrices of autoregressive coefficients for lag j , and u_t is the vector of reduced-form residuals. We assume that u_t is a white noise process with zero mean and covariance matrix $\Sigma_u = E[u_t u_t']$.

The model is estimated equation-by-equation using **Ordinary Least Squares (OLS)**. Given the lag length of $p = 4$, the estimation is performed on the effective sample starting from the 5th observation to account for the initial conditions.

3.2 Inference: The Residual Bootstrap

To construct reliable confidence intervals for the Impulse Response Functions (IRFs), I employ a recursive **residual-based bootstrap** procedure. This approach is preferred over asymptotic approximation, especially in finite samples, as it does not rely on strict normality assumptions for the error terms.

The bootstrap algorithm is implemented as follows:

1. First, the VAR parameters $(\hat{A}_j, \hat{c})$ and the residuals $\hat{u}_t$ are estimated from the actual data via OLS.

2. A set of bootstrap residuals $\{u_t^*\}$ is generated by randomly resampling with replacement from the centered empirical residuals $\hat{u}_t$.
3. Artificial time series data $\{y_t^*\}$ are constructed recursively using the estimated coefficients and the resampled residuals:

$$y_t^* = \hat{c} + \sum_{j=1}^p \hat{A}_j y_{t-j}^* + u_t^* \quad (2)$$

starting from the initial conditions of the original sample.

4. The VAR model is re-estimated on the artificial data $\{y_t^*\}$ to obtain a new set of coefficients $\hat{A}^*$ and covariance matrix $\hat{\Sigma}_u^*$.

This process is repeated $N_{boot} = 200$ times. For each replication, I compute the impulse responses and identifying restrictions. The 16th and 84th percentiles of the resulting distribution are used to construct the 68% confidence bands reported in the results.

4 Methodology: Structural Identification

4.1 From Reduced-Form to Structural Shocks

The reduced-form residuals u_t obtained from the OLS estimation represent unexpected movements in the variables, but they are contemporaneously correlated and do not carry a specific economic meaning. To perform policy analysis, I must recover the orthogonal *structural shocks* ε_t , which are assumed to be uncorrelated with unit variance ($\varepsilon_t \sim N(0, I_n)$).

The relationship between reduced-form residuals and structural shocks is given by:

$$u_t = B\varepsilon_t \quad (3)$$

where B is the structural impact matrix. The identification problem arises because there are infinitely many matrices B that satisfy the covariance constraint $\Sigma_u = BB'$.

To solve this, I employ a method based on **rotation matrices**. I assume that the impact matrix takes the form $B = PQ$, where P is the Cholesky factor of Σ_u (such that $PP' = \Sigma_u$) and Q is an orthonormal rotation matrix ($QQ' = I$). The identification challenge is to find a specific rotation matrix Q that yields impulse responses consistent with economic theory.

4.2 Mixed Sign Restrictions Approach

Unlike standard recursive identification (Cholesky), which imposes a rigid ordering of variables, I adopt a **Mixed Sign Restrictions** approach. This method combines:

- **Zero Restrictions (0):** Used to enforce the assumption of "sluggishness" for real and nominal variables that react slowly to financial shocks.

- **Sign Restrictions** (< 0 or > 0): Used to characterize the immediate direction of financial variables and monetary policy.

Specifically, I search for a rotation matrix Q such that the resulting impulse response functions (IRFs) at the impact period ($t = 0$) satisfy the conditions outlined in Table 1.

Variable	Restriction ($t = 0$)	Economic Logic
1. Oil Price	0	<i>Exogeneity</i> : UK shocks do not affect global oil prices.
2. Ind. Production	0	<i>Real Rigidity</i> : Output takes time to adjust to financial shocks.
3. Inflation	0	<i>Nominal Rigidity</i> : Prices are sticky in the very short run.
4. Unemployment	0	<i>Labor Rigidity</i> : Hiring and firing decisions are not instantaneous.
5. Policy Rate	< 0	<i>Monetary Policy</i> : Immediate rate cut to support the economy.
6. Exchange Rate	< 0	THE SHOCK : Immediate depreciation (Sterling crash).

Table 1: Identification Restrictions for the Brexit Depreciation Shock.

4.3 Justification of the Restrictions

The logic behind this identification scheme is twofold:

1. Sluggishness of the Real Economy (The Zeros). I assume that the "real economy" (Production and Unemployment) and the "nominal economy" (Inflation) react with a lag of at least one month to financial shocks.

- **Oil Price (0)**: As a small open economy, the UK is a price-taker in the global oil market. A domestic depreciation shock cannot contemporaneously influence the global price of Brent crude. This zero restriction is crucial to disentangle the Brexit shock from global supply shocks (e.g., the 2008 crisis).
- **Unemployment (0)**: Labor market adjustments involve search and matching frictions. Firms cannot instantly lay off or hire workers in response to a sudden exchange rate movement within the same month.

2. Simultaneity of Financial Markets (The Signs). In contrast to the real economy, financial markets and the central bank can react instantaneously.

- **Exchange Rate (< 0)**: This restriction defines the shock itself. I look specifically for a structural break that causes a sudden devaluation of Sterling, consistent with the events of June 2016.

- **Policy Rate (< 0):** This is the key improvement over Cholesky identification. I allow the Bank of England to react *within the same month* to the currency crash by cutting rates (or keeping them low). A recursive Cholesky structure would incorrectly force this response to zero, ignoring the simultaneous nature of financial and monetary decisions.

Implementation Note: The algorithm draws random rotation matrices Q . If a draw produces IRFs that satisfy all the conditions in Table 1 at $t = 0$, the draw is kept; otherwise, it is discarded. This process is repeated until 1,000 valid models are identified to construct the confidence bands.

5 Empirical Results

5.1 Impulse Response Analysis

Figure 1 presents the impulse response functions (IRFs) of the six endogenous variables to the identified Brexit depreciation shock over a horizon of 48 months. The solid black line represents the median response, while the dashed red lines indicate the 68% confidence intervals generated via bootstrapping ($N_{boot} = 200$).

The dynamic adjustments of the UK economy can be summarized as follows:

1. **The Shock (Exchange Rate & Policy Rate):** By construction, the Exchange Rate depreciates significantly on impact ($t = 0$). The persistence of this shock is remarkable: Sterling remains below its pre-shock level for the entire 4-year horizon, confirming the structural nature of the break. Simultaneously, the Shadow Policy Rate falls and remains negative, capturing the Bank of England's accommodative stance (rate cuts) aimed at mitigating the crisis.
2. **Inflation (Pass-Through):** We observe a rapid, albeit volatile, positive response in CPI Inflation. This is consistent with the "Exchange Rate Pass-Through" theory: a weaker currency increases import prices, which feeds into consumer inflation, reducing real household income.
3. **Unemployment (The Key Finding):** The response of the Unemployment Rate provides the central insight of this study. Following the shock, unemployment does *not* fall. Instead, it exhibits a statistically significant **hump-shaped increase**, peaking around 12-18 months after the shock.

This result strongly supports the **Uncertainty/Demand Hypothesis**. Despite the theoretical possibility that reduced EU migration (Labor Supply Shock) could tighten the labor market, the data shows that recessionary forces dominated. The combination of uncertainty, lower investment (reflected in the initial dip in Industrial Production), and the squeeze on real incomes led firms to freeze hiring, causing unemployment to rise relative to the baseline.

Impact of Brexit Depreciation Shock on UK Economy

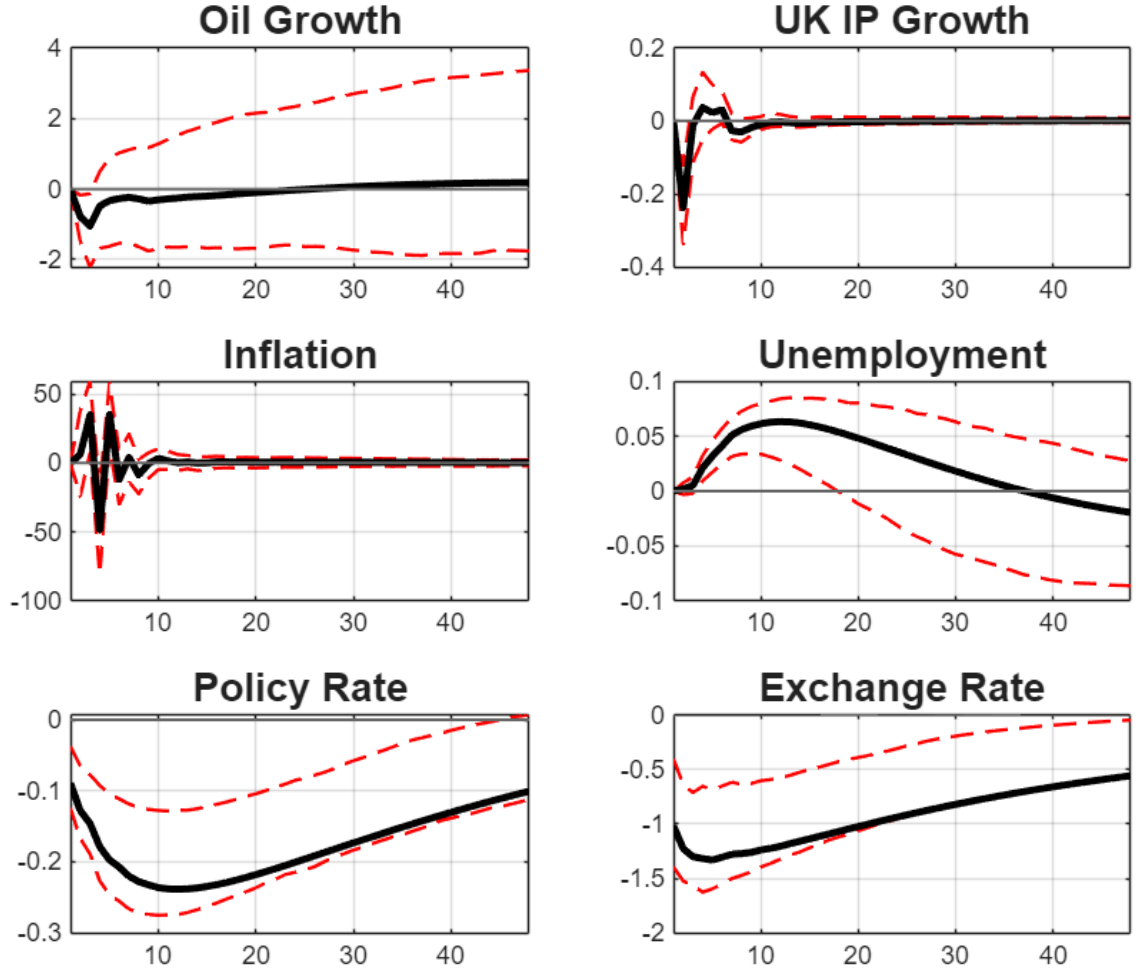


Figure 1: Impulse Responses to a Brexit Depreciation Shock. Solid line: Median; Dashed lines: 68% Confidence Bands.

5.2 Variance Decomposition

To quantify the relative importance of the Brexit shock compared to other macroeconomic drivers, Figure 2 reports the Forecast Error Variance Decomposition (FEVD) at horizons of 12, 24, and 48 months.

The analysis reveals that the exchange rate shock plays a relevant role in the short-to-medium term. Specifically, the shock explains approximately **10%** of unemployment fluctuations at the 12 and 24-month horizons. However, its explanatory power declines over the longer run, dropping to around **5%** at the 48-month horizon.

This pattern suggests that the Brexit depreciation was a significant driver of labor market dynamics during the first two years post-referendum (the medium term), but its relative

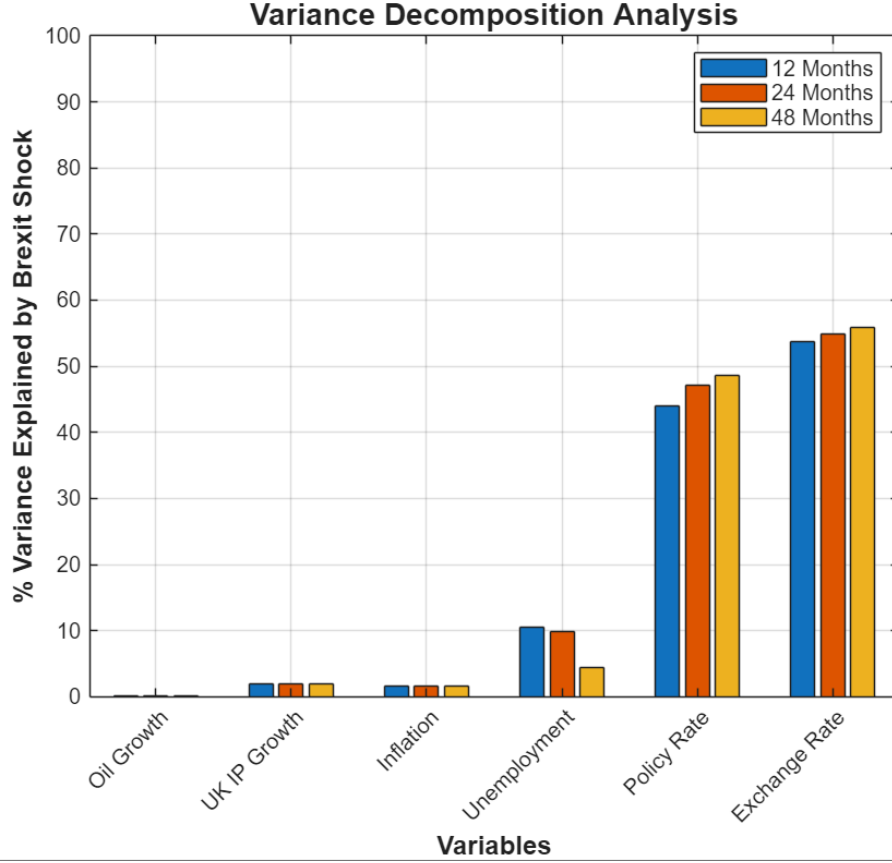


Figure 2: Forecast Error Variance Decomposition: Percentage of variance explained by the Brexit Shock.

influence fades in the long run as other structural factors or shocks become dominant.

5.3 Robustness Check: Comparison with Cholesky

To validate the necessity of our Mixed Sign Restrictions, we compare our results with a standard recursive identification (Cholesky decomposition), where the Exchange Rate is ordered last. Figure 3 illustrates this comparison for the Policy Rate and Unemployment.

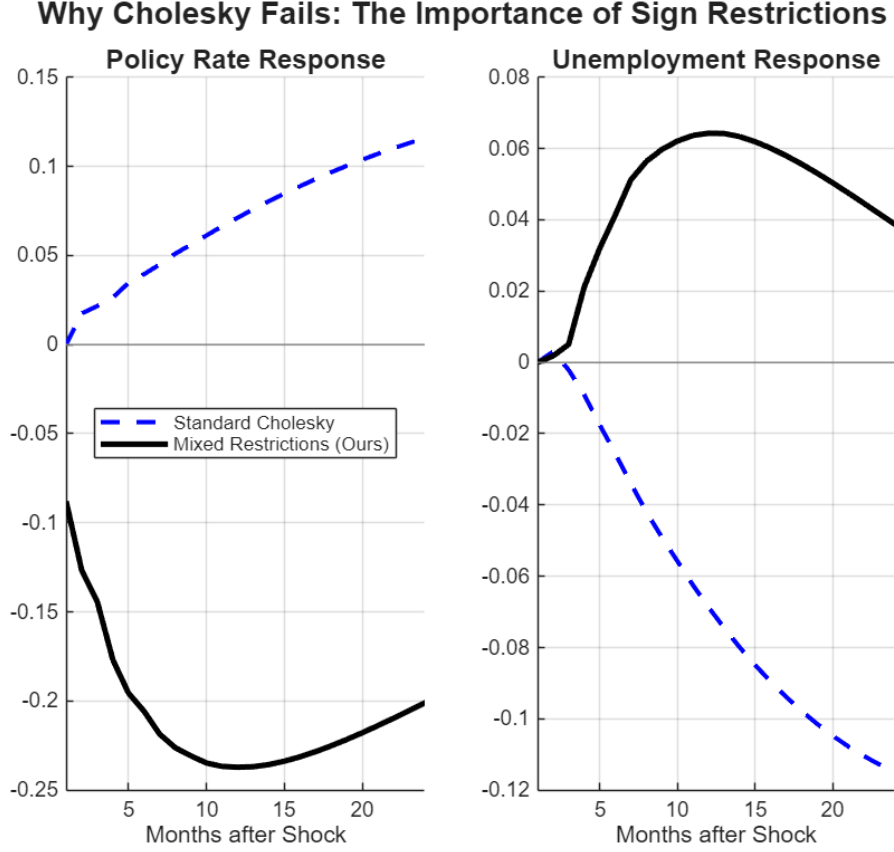


Figure 3: Methodological Comparison: Mixed Sign Restrictions (Black/Solid) vs. Standard Cholesky (Blue/Dashed).

The comparison highlights a critical bias in the standard approach:

- **Policy Rate Bias:** The recursive Cholesky model (blue dashed line) fails to capture the simultaneous monetary policy accommodation at $t = 0$, erroneously implying a zero or positive interest rate response.
- **Unemployment Bias:** Consequently, the Cholesky model underestimates the recessionary impact, suggesting a spurious decline in unemployment (implying that the export-competitiveness channel dominates).

Our Mixed Sign approach corrects this identification failure, revealing the true recessionary nature of the shock (unemployment increase) by properly accounting for the simultaneous reaction of financial variables.

It is important to emphasize that the failure of the Cholesky identification demonstrated in Figure 3 is not merely a result of the specific ordering chosen (Exchange Rate last), but rather an inherent limitation of recursive schemes in the presence of financial variables.

One might argue that placing the Exchange Rate *before* the Policy Rate would allow the Central Bank to react contemporaneously. However, this alternative ordering would simply reverse the bias rather than eliminate it. By placing the Exchange Rate first, the model would impose that the Exchange Rate cannot react contemporaneously to a monetary policy shock

within the same month. This assumption is equally unrealistic in highly liquid financial markets, where asset prices respond instantly to central bank announcements.

Therefore, regardless of the relative ordering of financial variables, Cholesky identification forces a "unidirectional causality" (either $Exchange \rightarrow Rate$ or $Rate \rightarrow Exchange$) that is structurally incapable of capturing the ****simultaneity**** typical of a financial crisis. The Brexit shock was characterized by a mutual and instantaneous interaction between the currency crash and the monetary policy response. Only the **Mixed Sign Restrictions** approach, by eschewing zero restrictions on the impact matrix for these variables, can correctly model this bidirectional feedback loop.

6 Conclusion

This paper has investigated the dynamic macroeconomic effects of the Brexit-induced depreciation of the Sterling on the UK labor market. By employing a **Structural Vector Autoregression (SVAR)** model identified via **Mixed Sign Restrictions** on monthly data from 2000 to 2024, I successfully isolated the exogenous exchange rate shock from other global and domestic disturbances.

The empirical analysis yields three major conclusions that clarify the nature of the post-referendum economic adjustment:

1. **Dominance of the Uncertainty Channel:** The central finding of this study is that the Brexit depreciation shock generated a **statistically significant and persistent increase in the unemployment rate**, which peaked approximately 12 to 18 months after the shock. This evidence strongly supports the "Uncertainty/Demand Hypothesis," suggesting that the spike in economic policy uncertainty and the erosion of real incomes (via imported inflation) acted as a severe negative demand shock, inducing firms to freeze hiring and investment.
2. **Refutation of Expansionary Hypotheses:** Crucially, our results allow us to jointly refute two popular theoretical channels that predicted a tightening of the labor market (i.e., lower unemployment) following the vote:
 - *The Competitiveness Channel:* Contrary to standard trade theory, the massive depreciation did not stimulate labor demand enough to lower unemployment via an export boom.
 - *The Labor Supply Channel:* The anticipated reduction in EU migration did not lead to a "scarcity of labor" effect strong enough to reduce unemployment.

The data indicates that these potentially expansionary forces were completely overwhelmed by the contractionary demand shock. The fear of future trade barriers and lower profitability proved to be the dominant driver of labor market dynamics.

3. **Methodological Insights & Policy Limits:** Finally, the analysis highlights the limitations of standard econometric tools and monetary policy. First, the comparison with recursive identification (Cholesky) demonstrates that accounting for the simultaneous interaction between the exchange rate and interest rates is vital; neglecting it would have biased the results towards a spuriously benign outcome. Second, while the Bank of England reacted promptly by lowering the policy rate, this accommodation was insufficient to fully offset the recessionary forces unleashed by the structural break.

In summary, the Brexit event acted primarily as a persistent negative shock to the UK economy. Far from acting as a lever for competitiveness or wage growth via labor scarcity, the depreciation and associated uncertainty imposed a significant adjustment cost on the labor market in the medium term.

Data Sources

I constructed the dataset using monthly series retrieved from the FRED database (Federal Reserve Bank of St. Louis) and the Bank of England. Below are the direct links to the series used:

Global Oil Price: Brent Crude Oil Price.

<https://fred.stlouisfed.org/series/POILBREUSD>

Industrial Production: Production of Total Industry in the United Kingdom.

<https://fred.stlouisfed.org/series/GBRPROINDMISMEI>

CPI Inflation: Consumer Price Index: All Items for United Kingdom.

<https://fred.stlouisfed.org/series/CPALTT01GBM659N>

Unemployment Rate: Harmonized Unemployment Rate: Total: All Persons for the UK.

<https://fred.stlouisfed.org/series/LRHUTTTTGBM156S>

Policy Rate: Bank of England Official Bank Rate.

<https://fred.stlouisfed.org/series/INTDSRGBM193N>

Exchange Rate: Real Effective Exchange Rate for United Kingdom.

<https://fred.stlouisfed.org/series/RBGBBIS>